IFT6163: Project Proposal

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Abstract

This semester, for our project, we explore the benefits of training a policy for a robot in a simulation and transfering that policy to a real world physical robot. Our goal is to understand the benefits of using `Sim2Real` training for a policy as well as its negatives on said training. Afterwards, we want to explore the benefits that instilling intrinsic motivation into our robot's training could bring. We hope that training our policy with some intrinsic motivation built into it will increase behavioral diversity in simulation, increasing the robustness of the policy trained in simulation when transfered to physical hardware.

1 [4 pts] Project Method, How will the method work (2-3 paragraphs + figure(s)/math)

The methodology we propose utilizes a **comparative experimental design** to evaluate the efficacy of self-supervised exploration in bridging the "reality gap" for navigation task. We investigate how a policy trained with a "curiosity" signal developped through intrinsic rewards generalizes to physical hardware compared to a baseline agent trained via extrinsic rewards.

The primary training environment will be developped in **PyBullet**. We will utilize **Stable Baselines3** as our reinforcement learning framework, specifically the **Proximal Policy Optimization (PPO)** algorithm.

To facilitate the **Sim2Real** transfer, we implement **Domain Randomization (DR)** (Tobin et al. [2017]). During training, we will stochastically vary physical parameters such as friction to ensure the policy doesn't overfit to a "perfect" and unchanging environment.

Our baseline PPO agent will optimize a purely **Extrinsic Reward** (r_{ext}) policy. To test our hypothesis, we will modify the reward function to include **Intrinsic Rewards** (r_{int}). The total reward at time step t will be:

$$r_t = r_{ext} + \beta \cdot r_{int}$$

Where β is a decay coefficient to ensure that the robot isn't stuck in exploration mode during the entirety of training, but eventually shifts it's focus towards completing the task we give it.

In order to instill curiosity into our model, we will use the **ICM** architecture featured in the paper published by Pathak et al. [2017].

2 [2 pts] What is the related work? Provide references: (1 paragraph)

Reinforcement learning (RL) has become a standard approach for learning control policies directly from interaction, Policy gradient methods such as Proximal Policy Optimization (PPO) Schulman et al. [2017] are widely used due to their stability and simplicity in continuous control settings. However, training purely from extrinsic task rewards can be data inefficient, especially in sparse-reward environments. To address this, several works have introduced intrinsic motivation mechanisms that encourage exploration and add signal throughout the training, such as UNREAL Jaderberg et al. [2016] and Intrinsic Curiosity Modules (ICM) Pathak et al. [2017].

In parallel, Sim2Real transfer has been extensively studied as a way to leverage simulation for scalable training, but closing the gap between simulated and real-world dynamics remains a central challenge. Our project builds on these lines of work by studying whether intrinsic motivation during simulation training can improve learning efficiency and increase robustness and zero-shot transfer performance when deploying policies to real robotic hardware.

3 [2 pts] What resources will you use (e.g. software) or need (e.g. hardware) and why: (1 paragraph)

Hardware: We will utilize the PiCrawler robot from **Sunfounder**.This robot is equipped with an intergrated camera and is a four-legged robot with 12 servos. Currently, we are looking for a **URDF** or **CAD** file that would allow us to simulate the robot inside of our simulation environment. From what we have seen, contacting the **Sunfounder** team could solve this. In the unfortunate event where we are unable to obtain such a file for this specific robot or a file for a similarly built robot, we will have to pivot to another robot of the same vein.

Software & Compute: For this project, we will use the **PyBullet** simulation environment for all of our simulation training. We will be leveraging the **Stable Baseline3** python library for our **RL** algorithm implementation. Given that we will be training a policy in a simulation environment, we will require a decent amount of compute. While it would be highly unideal, we can obtain that compute from Google Colab. However, having access to a computer with a sufficiently performant NVIDIA GPU or a compute server with enough memory would be the prefered option and would likely suffice for this project.

4 [2 pts] What new skills you(s) will learn from this project?

Throughout this project, we hope to cultivate a set of technical skills that would allow us to bridge the gap between the theory seen in class and it's practical applications on physical hardware. We mainly hope to expand our knowledge in these following areas:

- **Reinforcement Learning:** This project will allow us to have hands on experience in training and using **RL** policies. This includes understanding and mastering the intricacies of reward shapping and policy optimization.
- **Sim2Real Transfer:** We will develop the ability to navigate the reality gap that simulation training can incur. This includes learning how to minimize the reality gap by calibrating sensors and other parts of the physical robot to ensure that the policy remains robust during the transfer phase and stays optimal.
- **Implementation of Intrinsic Motivation:** We will learn to implement intrinsic motivators to our robots policy during training. This will allow us to deepend our understanding of how curiosity-driven training can impact the performance of the policy.
- **Hardware Deployment:** In this project, we will learn to deploy a trained policy on a physical robot and test it inside of a real environment. This will allow us to understand the limitations of simulation testing as well as it's downfalls.

By tackling all of these challenges, we hope to go beyond the theory seen in class and put it in practice on real physical hardware. Ultimately, this deep dive into these subjects will allow us to gain a deeper understanding of an unfamiliar field to both of us.

5 [2 pts] What is the timeline for the project? At least month-by-month granularity, ending with a complete project (8-12 pages with figures).

1. **Feburary**
   (a) Review of the literature
   (b) Establish and divide implementation responsibilities
   (c) Repository setup
   (d) Implement and validate simulation environment
2. **March**
   (a) Implement intrinsic reward method
   (b) Train a baseline policy in simulation using extrinsic rewards only
   (c) Train a second policy in simulation using a combination of extrinsic and intrinsic rewards
   (d) Compare learning curves and exploration behavior in simulation
   (e) Transfer baseline policy to physical hardware and evaluate performance
3. **April**
   (a) Conduct real-world evaluations
   (b) Compare Results
   (c) Write report

References

Max Jaderberg, Volodymyr Mnih, Wojciech Marian Czarnecki, Tom Schaul, Joel Z Leibo, David Silver, and Koray Kavukcuoglu. Reinforcement learning with unsupervised auxiliary tasks. 2016. URL https://arxiv.org/abs/1611.05397.

Deepak Pathak, Pulkit Agrawal, Alexei A. Efros, and Trevor Darrell. Curiosity-driven exploration by self-supervised prediction. 2017.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. 2017. URL https://arxiv.org/abs/1707.06347.

Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world, 2017. URL https://arxiv.org/abs/1703.06907.